CHAPTER IV

RESULTS AND DISCUSSION

4.1 Implementation

This chapter presents the results of the development and implementation of the Gonzaga Dental Clinic Management Automation System (GDCMAS). It discusses the development tools used, hardware and software requirements, inputs and outputs of the study, results interpretation and analysis, and the system evaluation results.

4.1.1 Development Tools

Tool	Purpose
Visual Studio Code	Integrated Development Environment (IDE) for coding and debugging
Git and GitHub	Version control and repository management
Node.js and npm	JavaScript runtime and package management
Next.js 16	Frontend and backend framework for web application development
TypeScript	Programming language for type-safe development
Tailwind CSS 4	CSS framework for UI design and styling
Supabase	Backend-as-a-Service providing PostgreSQL database, authentication, and storage

4.1.2 Hardware Requirements

For Development

Component	Minimum Requirement	Recommended
Processor	Intel Core i5 or equivalent	Intel Core i7/AMD Ryzen 7 or higher
RAM	8 GB	16 GB or higher
Storage	256 GB SSD	512 GB SSD or higher
Internet Connection	Stable internet connection	Stable high-speed internet connection

4.1.3 Software Requirements

Software	Purpose

Operating System	Windows 10/11, Linux, or macOS
Web Browser	Google Chrome, Mozilla Firefox, or Microsoft Edge (Latest Version)
Node.js	Version 18.18.0 or higher
Supabase	Cloud-based PostgreSQL database and authentication service
Git	Version control system

4.2 Input and Outputs of the Study

4.2.1 Inputs

The system requires the following inputs from authorized users: Patient registration details, Medical history, allergies, and medications, Dental treatment information, Appointment details, Consent information, Staff account details, and Patient booking requests.

4.2.2 Outputs

The system generates: Patient records and profiles, Treatment history reports, Appointment schedules per branch, Daily/branch overview dashboards, Consent documentation, Audit logs, Public booking availability and confirmation messages, and Staff management reports.

4.3 Results Interpretation and Analysis

The implementation of GDCMAS addressed the identified problems. Centralized patient records, improved appointment management, enhanced security and access control, efficient patient booking, and accountability through audit logging were achieved.

4.4 System Evaluation Results

Module	Evaluation Criteria	Result	Remarks
Patient Management	Registration, search, duplicate detection	Passed	Successfully manages patient records with duplicate prevention
Treatment Records	Tooth chart, procedures, history tracking	Passed	FDI tooth chart implemented with complete treatment documentation
Appointment Scheduling	Slot generation, booking, status management	Passed	Efficient scheduling with cross-branch conflict prevention
Public Booking	Availability, request submission	Passed	Functional unauthenticated booking portal
Staff Management	Roles, permissions, activation	Passed	Role-based access control properly implemented
Security & RLS	Data isolation, access control	Passed	Multi-layered security successfully enforced
Audit Logging	Action tracking	Passed	All critical actions are logged

4.5 Discussion

The GDCMAS successfully achieves its goal of automating the Gonzaga Dental Clinic's manual processes. The implementation of Row Level Security ensures that sensitive patient data remains

protected while still allowing necessary access across branches for continuity of care.